HW#11

1. $X(k) = [-2,\ 1+j,\ 3,\ 1-j,\ 2]$

(a) $x(-n)_N$

since $x(-n)_N \longleftrightarrow X(-k)_N$

$$\begin{array}{l|l} & -2,\ 1+j,\ 3,\ 1-j,\ 2 \\ 2,\ 1-j,\ 3,\ 1+j, & -2,\ 2,\ 1-j,\ 3,\ 1+j,\ -2 \end{array}$$

so that the DFT of $x(-n)_N$ is $[-2,\ 2,\ 1-j,\ 3,\ 1+j]$

(b). Based on the Parseval's theorem,

$$\sum_{n=0}^{4} |x(n)|^2 = \frac{1}{5}\sum_{k=0}^{4} |X(k)|^2 = \frac{1}{5}[4+4+2+9+2] = 4.2$$

(c). $\sum_{n=0}^{4} |x(n-2)_N|^2 = \sum_{n=0}^{4} |x(n)|^2| = 4.2$.

2. proof: $W_N = e^{-j\frac{2\pi}{N}}$, $W_{\frac{N}{4}}^{k} = \left[e^{-j\frac{2\pi}{N/4}}\right]^k = e^{-j\frac{8\pi k}{N}} = e^{-j\frac{2\pi(4k)}{N}} = W_N^{4k}$

3. $F_S = 16K\text{Hz}$, $t = 0.4s$, key 5, with $f_1 = 770\text{Hz}$ and $f_2 = 1336\text{Hz}$

$N = \frac{t}{T_S} = t \cdot F_S = 0.4 \times 16K = 6400$

since $\Omega_k = \frac{2\pi k}{N} \Rightarrow \omega_k = \frac{2\pi k}{NT_S} \Rightarrow f_k = \frac{k}{NT_S} = \frac{k}{N} \cdot F_S$

so that $k = \frac{f_k \cdot N}{F_S} \Rightarrow k_1 = \frac{770 \times 6400}{16 \times 10^3} = 308$, $k_2 = \frac{1336 \times 6400}{16 \times 10^3} = 534$

4. (a) the duration of the signal: $t = N \cdot T_S = N \cdot \frac{1}{F_S} = 7200 \cdot \frac{1}{12 \times 10^3} = 0.6s$

(b) resolution of analog frequency. $\Delta f = \frac{\Delta\omega}{2\pi} = \frac{\frac{\Delta\Omega}{T_S}}{2\pi} = \frac{\frac{2\pi}{NT_S}}{2\pi} = \frac{F_S}{N} = \frac{12 \times 10^3}{7200} = 1.67\text{Hz}$

(c). $f_1 = \frac{k_1}{N} \cdot F_S = \frac{512}{7200} \times 12 \times 10^3 \doteq 853\text{Hz}$

$f_2 = \frac{k_2}{N} \cdot F_S = \frac{803}{7200} \times 12 \times 10^3 \doteq 1338\text{Hz}$

(d) Based on the DTMF table, the signal is generated by key '8'.